

THE CHAI STRAINER AND THE SALT PAN

Learning to Separate Magic Mixtures!

GRADE 4 SCIENCE LESSON HANDBOOK

Kabir and Anya have a muddy playground problem!

Let's explore how scientists separate messy mixtures using smart
tricks!

Kabir and Anya's Rainy Day Challenge

Oh no! A huge rainstorm just hit the neighborhood playground!

After a heavy, splashing rain, Kabir and Anya's favorite cricket ground was turned into a total muddy mess! While trying to clean up, their water bucket fell over and got completely mixed up with dirt, brown sand, and salt.

Now they are stuck with a bucket of dirty, salty, muddy brown liquid. They need our help!

The Big Question:

Can you help Kabir and Anya separate this big mess to get clean water, clean sand, and their salt back? Let's learn the secret methods together!

What is a Mixture?

Before we can fix the bucket, we need to know what happened inside it. When you put things together, you create a **mixture**!

A mixture happens when two or more different things are mixed up, but they can still be taken apart or separated back into their original forms.

Let's look at some daily examples of mixtures:

- Tea leaves floating around in hot water.
- Brown sand mixed up in a glass of water.
- White sugar or salt stirred into warm water.

Two Crucial Science Words:

SOLUBLE

Something that can dissolve (melt completely) into liquid and disappear, like salt!

INSOLUBLE

Something that cannot dissolve, so it floats around or sinks, like sand!

How Filtration Works: The Barrier Rule

Our very first trick for taking mixtures apart is called **Filtration**. It works by using a barrier with tiny, microscopic holes!

Imagine a tiny net or a screen. When you pour a mixture through it, the fluid flows right through the holes, but solid blocks that are too big get stopped completely!

The Residue (The Leftovers)

The solid bits that are too big to pass through the filter get trapped and stay behind. This trapped solid is called the **Residue**!

The Filtrate (The Clean Liquid)

The clean liquid that successfully passes through the tiny holes in the filter is called the **Filtrate**!

An Indian Example: The Chai Strainer

You see filtration happen every single day in your kitchen!

When your parents make hot morning tea, they use a special tool called a "**Chai-ni**" (a tea strainer). This strainer has a net pattern made of tiny gaps.



Inside the Kitchen Lab:

When the messy tea is poured into the cup:

- The strainer stops the big black tea leaves from falling into your cup. This trapped pile is the **Residue**!
- The delicious, smooth liquid tea passes right through the net gaps. This clean liquid is our **Filtrate**!

So, a Chai-ni is just a brilliant science filter!

How Evaporation Works!

But wait! What happens if a solid is soluble and dissolves completely, like salt? The salt particles become too tiny to be caught by a filter cloth. They slip right through the holes with the water!

To catch dissolved pieces, we use a different magic trick called **Evaporation**! We use heat instead of a filter cloth.

The Disappearing Act:

When salty water is heated, the liquid water transforms into invisible steam and floats away into the sky. But the salt particles cannot turn into steam! They get left behind at the bottom of the pan as dry white crystals.

 **Safety Note: Heating experiments can be dangerous. Always do heating activities with an adult helper watching you!**

An Indian Example: The Salt Pan

Did you know that the salt you eat on your food is made by the sun?

In sunny coastal places like Gujarat and Tamil Nadu, salt farmers use natural evaporation on a giant scale! They build huge, flat, shallow ground spaces called **salt pans**.

Sun Powered Science:

First, they let salty seawater flow into these wide pans. Then, they let the blazing hot sun heat the water day after day. As the water disappears into steam up into the blue sky, beautiful white salt crystals are left behind on the ground!

The salt farmers simply rake up these leftover crystals to give us our kitchen salt!

When We Need Both Magic Steps!

Now we have all the super science tools to solve Kabir and Anya's muddy water challenge! Because their bucket has both insoluble sand and soluble salt, we must use both methods in order!

The Action Plan

STEP	WHAT HAPPENS?
STEP 1: Filtration	Pour the mixture through a filter cloth. The heavy sand gets TRAPPED on top (Residue). The clear salt water flows right through (Filtrate).
STEP 2: Evaporation	Heat that clear salt water filtrate gently in a flat pan. The water turns to steam and disappears into the air, leaving clean SALT CRYSTALS behind!

Hooray! The bucket is clean, and all elements are separated safely!

Guided Lab Practice: River Water Cloth Filter

Let's look at how Kabir uses this science at home. Kabir folds a clean piece of *Suti Kapda* (white cotton cloth) and places it tightly over an empty bucket. He slowly pours muddy river water through it...

Let's identify what happens inside the lab:

- **What gets TRAPPED in the cloth?** Silt (fine mud particles), dirt, small pebbles, and tiny leaf bits. This is our **Residue**!
- **What PASSES THROUGH the cloth?** Clear, cleaner water is collected down below. This is our **Filtrate**!

The Rann of Kutch Fact:

The famous Rann of Kutch in Gujarat is one of the world's largest salt deserts! It works exactly like a giant evaporation lab run by the hot sun!

Summary: Remember Your Rules!

- **Insoluble? Filter It!** If a solid cannot dissolve, filter it to catch it as **Residue**.
- **Soluble? Evaporate It!** If it dissolves like salt, heat it so water turns to steam and leaves **Crystals** behind.

Your Secret Home Mission!

Watch an adult make hot chai tea or filter water at home. Be a science detective!

- Spot the **Residue** trapped behind (like black tea leaves).
- Spot the **Filtrate** passing into the cup (smooth liquid tea).
- Spot the **Evaporation** steam rising from the hot pot!

Quick Check: Can salt be removed by a cloth filter?

No! It is dissolved and needs heat!

THANK YOU! KEEP EXPLORING!

Student Practice Workbook (Grade 4)

Time to test your secret science superpowers! Answer the questions below based on what you learned in the lesson.

1. What is a mixture? Give one everyday example that you can find in your kitchen.

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2. If a substance completely dissolves in water and disappears, is it called soluble or insoluble?

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3. Imagine you mix small plastic beads with water. Will the beads be soluble or insoluble? Why?

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4. What is the science name given to the solid pieces trapped inside a filter screen?

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5. When you strain hot tea using a Chai-ni, what is the clear liquid called that collects in your cup?

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6. Why can't we separate dissolved salt from water by using a simple cotton cloth filter?

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7. What is the process called where liquid water turns into invisible steam due to heat?

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8. How do salt farmers in Gujarat and Tamil Nadu separate salt from seawater? Which natural resource heats the water?

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9. If you have a mixture of sand and water, which method is best to get the dry sand back quickly?

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10. In Kabir's muddy playground problem, why did he have to do filtration before doing evaporation?

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11. What huge salt desert in India functions like a giant natural evaporation lab?

12. True or False: When water evaporates from a saltwater mixture, the salt also turns into steam and flies into the sky.

13. Draw lines or explain what happens to the water and the tea leaves when making Chai. Which is the residue and which is the filtrate?

14. Give one safety rule you must remember whenever doing a science activity involving heat or evaporation.

15. You are given a mixture of iron paperclips, sand, and sugar. How could you separate them? (Hint: Think about magnets first, then water!)
